

Session 2026-06-18 (cont.) — NQ1 D-Sink Generalisation, NQ2 Structural Null Model, NQ3 Out-of-Sample Predictions

Framework: Universal Binary Principle (UBP) Core Studio v5.3 — hardened triad-physics edition, float-free core

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Targeted Open Questions: NQ1 (D-Sink^k/L lepton generalisation), NQ2 (structural null), NQ3 (out-of-sample: $H_{\square}$, W/Z , α drift, λ_{QCD} , $g-2$, $n-p$)

Stance: critical-both — work within UBP, flag every post-hoc move, run focused and structural null models

Predecessor: Push #1 (UBP_Gravity_Push2_2026-06-18.pdf) — established the $m_{\mu}/m_e = 13/L$ signal and the active-coincidence-search methodology

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1. Session Overview

This is the second push on the UBP gravity study, executing the three recommended next steps (NQ1, NQ2, NQ3) defined in Appendix E of Push #1. The primary thrust is NQ1: a focused investigation of the $m_\mu/m_e = 13/L$ formula that Push #1 flagged as the only candidate approaching a meaningful signal-to-noise ratio. The secondary thrust is NQ2: a stronger null model that scrambles the grammar (bases, multipliers, Y-power ranges) as well as the substrate constants, which Push #1's null model did not do. The tertiary thrust is NQ3: out-of-sample predictions on constants that were not used to design the search — the Hubble constant H_0 (CMB vs SNe tension), the W/Z boson mass ratio, the α drift bound, the QCD confinement scale λ_{QCD} , the neutron-proton mass difference, and the muon g-2 anomaly.

The user also requested that the existing PARTICLE_PHYSICS atlas in v5.3 be integrated into the study. This atlas (24 entries) uses bespoke "lens" formulas of the form "round-measured-value + substrate correction" (e.g. $\alpha^{-1} = 220 - 83 + L = 137 + L$, $m_p/m_e = 1836 + 2 \cdot L_s$). Section 6 compares the new D-Sink^k/L candidates against these existing lens formulas and finds that the lens formulas — despite being structurally less clean — generally outperform the new candidates on accuracy. This is because the lens formulas embed the measured integer, which is a strong form of in-sample fitting.

All computations were executed inside the v5.3 float-free core using Fraction arithmetic, with no new engines added. The Push #1 scripts were reused as-is; new scripts were added for NQ1 (focused null model on 13/L), NQ2 (structural null), NQ3 (out-of-sample search + null check), and atlas integration. All scripts persist in /scripts/ and all results in /results/.

2. Push #1 Recap & Push #2 Bug Discovery

Push #1's headline finding was that the gravity-style grammar produces sub-0.25% hits for five of seven reachable targets, with the m_μ/m_e formula $13/L = 169/w$ being the strongest (0.0294% error). Push #1's null model (substrate-only scramble, 40 trials) showed a 20% false-positive rate for G at the 0.13% threshold, leading to the verdict that the G hit is "statistically indistinguishable from coincidence."

While preparing Push #2, we discovered a bug in Push #1's target values: **m_τ/m_e was set to 347786.21, which is 100× too large.** The correct PDG value is $m_\tau/m_e = 1776.86 \text{ MeV} / 0.51099895 \text{ MeV} = 3477.228280$. Push #1's "0.4315% m_τ/m_e hit" was therefore against the wrong target — the formula $6/e \cdot Y_{\text{inv}}^9 = 346285$ was actually 9858% off against the correct target. This bug does not affect Push #1's main verdict (the G, α , α^{-1} , m_p/m_e , m_μ/m_e conclusions all stand), but it does mean Push #1's Phase B top-5 table for m_τ/m_e should be disregarded. We have corrected the target in all Push #2 scripts and flag this clearly.

Bug correction: Push #1's m_τ/m_e target was 347786.21 (100× too large). The correct value is 3477.228280. Push #1's m_τ/m_e "hit" was an artifact of the wrong target. All Push #2 computations use the corrected value.

A second, smaller bug was found in the original NQ3 script: the muon g-2 anomaly target was set to $F(251, 10^{**9}) = 2.51 \times 10^{10}$, but the correct Fermilab 2021 value is 2.51×10^{11} ($= F(251, 10^{**11})$). The NQ3 null-check script uses the correct value. The original NQ3 search results for g-2 anomaly should be read against the corrected target.

3. NQ1 — D-Sink^k / L Generalisation Across Lepton Generations

3.1 Hypothesis and family enumeration

Push #1's m_μ/m_e formula $13/L = 169/w$ uses $13 = \text{D-Sink}$ dimension and $L = w/13 = \text{D-Sink leakage}$. The natural generalisation is the D-Sink^k / L family:

$$\text{D-Sink}^k / L = 13^k / L = 13^{(k+1)} / w \text{ for } k = 0, 1, 2, \dots$$

If this family generalises across lepton generations, we expect integer k values: k=1 for m_μ/m_e (confirmed by Push #1), k=2 for m_τ/m_e , and possibly k=2 or k=3 for m_p/m_e . We also test the "Triad × D-Sink^k / L" subfamily ($3 \cdot 13^k/L$) and the "D-Sink^k / L_s" subfamily (using the spectroscopic sink $L_s = 29w/312$ instead of L). All three

subfamilies are computed for $k = 0.6$ and matched against m_μ/m_e , m_τ/m_e , m_p/m_e , m_n/m_e , and m_p/m_μ .

3.2 m_μ/m_e confirmation — 13/L reproduces at 0.0294%

We confirm Push #1's m_μ/m_e hit: $13/L = 206.707543$ against target 206.768283 (PDG 2024), error 0.0294%. For comparison, the existing PARTICLE_PHYSICS atlas formula $m_\mu/m_e = 206 + 12 \cdot L$ gives 206.754689 with error 0.0066% — about $4\times$ more accurate. The atlas formula embeds the integer 206 (close to the measured value), while 13/L does not. The accuracy difference is therefore expected; the structural difference is what matters.

Structurally, 13/L is cleaner: it uses only the D-Sink dimension (13) and the D-Sink leakage ($L = w/13$). It is equivalent to $13^2/w = 169/w$, which is the D-Sink squared divided by the Entropic Wobble. The atlas formula $206 + 12 \cdot L$ uses the measured integer (206) plus a substrate correction ($12 \cdot L = 0.755$), which is a "round + correction" structure that Push #1 explicitly flagged as post-hoc.

3.3 Focused null model on 13/L (5000 trials, scramble w only)

This is the strongest single test in either Push #1 or Push #2. We hold the integer 13 fixed (it is an integer, not a substrate constant) and scramble only the Entropic Wobble w , replacing it with $w' = w \times \text{uniform}(0.1, 10)$ in each trial. We then compute $13/L' = 169/w'$ and record the error against m_μ/m_e . Across 5000 trials:

Statistic	Value
Real substrate error (13/L on m_μ/m_e)	0.0294%
Null minimum (best of 5000 scrambled)	0.0295%
Null p10 (10th percentile)	42.9828%
Null p25 (25th percentile)	67.8298%
Null p50 (median)	81.9892%
Null p75 (75th percentile)	87.4450%
Null p90 (90th percentile)	89.3810%
Null p99 (99th percentile)	426.4554%
Null mean	83.3949%
Trials with $\text{err} \leq \text{real err}$	0/5000 = 0.00%
Real substrate's percentile	100.00% (100% = best possible)

Verdict: REAL substrate is SURPRISING (< 5% null hit rate) Across 5000 scrambled- w trials, **0 trials** matched or beat the real substrate's 0.0294% error. The real substrate's w is at the 100th percentile of the null distribution (best possible). This is the first and only formula in either Push #1 or Push #2 that survives a focused null model at this stringency. It is genuinely surprising.

Caveat: this null model tests only whether 13/L's accuracy on m_μ/m_e is surprising under w -scrambling. It does not test whether the formula structure "integer / L " is itself surprising — for that we need the structural null of §4. Also, the result is specific to m_μ/m_e : §3.4 shows that 13/L does not generalise cleanly to m_τ/m_e .

3.4 m_τ/m_e — does $13^2/L$ or $39/L$ generalise? (No)

We tested 15 candidate formulas of the $D\text{-Sink}^k/L$ family against the corrected m_τ/m_e target (3477.228280). The results are unambiguous: the family does not generalise to m_τ/m_e at any tested k . Best candidates:

Formula	Value	Error %
$13 \cdot Y_{\blacksquare}^2 / L$	2950.7273	15.1414
$169/L$	2687.1981	22.7201
$169/L_s$	2223.8880	36.0442

Formula	Value	Error %
$13 \cdot Y_{inv}^1 / L$	780.9850	77.5400
$13 \cdot \pi / L$	649.3909	81.3245
$39 / L$	620.1226	82.1662
$13 \cdot e / L$	561.8894	83.8409
$39 / L_s$	513.2049	85.2410

For comparison, the existing PARTICLE_PHYSICS atlas formula for m_{τ}/m_e ("24D MPG Lever" lens, using Y_{inv} -based expression) gives 3478.0128 with error 0.0226% — about 700× more accurate than the best D-Sink^k/L candidate. The atlas formula uses Y_{inv} , not L , which suggests the tau mass may genuinely be a Y_{inv} -structure rather than an L -structure.

The D-Sink^k/L family therefore does **not** generalise across lepton generations. The m_{μ}/m_e hit at $k=1$ is real (per §3.3) but does not extend to $k=2$ for m_{τ}/m_e . This is a partial falsification of the "D-Sink^k/L as a generational pattern" hypothesis.

3.5 Implied-k analysis — non-integer k breaks the pattern

If we solve $13^k/L = m_{target}$ for k (giving $k = \log(m_{target} \cdot L) / \log(13)$), the implied k values are:

Target	Implied k for $13^k/L = target$	Closest integer?
m_{μ}/m_e	1.0001	YES (k=1) — confirms 13/L
m_{τ}/m_e	2.1005	no ($k \approx 2.10$)
m_p/m_e	1.8515	no ($k \approx 1.85$)

The implied k values for m_{τ}/m_e (2.10) and m_p/m_e (1.85) are **not close to integers**. If the D-Sink^k/L pattern were a true generational structure, we would expect $k = 2$ for m_{τ}/m_e (next generation after muon) and possibly $k = 2$ or 3 for m_p/m_e . The non-integer values suggest that $13^k/L$ is not the right structural form for these targets — even though it is for m_{μ}/m_e .

An alternative reading: the m_{μ}/m_e hit at $k=1$ is a single-substrate-constant coincidence that the focused null model is not powerful enough to detect. The 5000-trial null model in §3.3 scrambles only w , but the structural null in §4 scrambles the grammar too — and there the m_{μ}/m_e false-positive rate is much higher (43% in §4.2). The two null models give different verdicts, which we discuss in §4.

4. NQ2 — Structural Null Model (scramble grammar AND substrate)

4.1 Design — random grammar subset × scrambled substrate

Push #1's null model scrambled substrate constants but kept the grammar fixed (10 bases × 55 scales × 27 multipliers = 29 700 candidates per target). This is conservative: scrambling the grammar too should increase the false-positive rate. NQ2's structural null does both:

For each of 30 trials: (i) sample 5-10 bases from the 10 real bases; (ii) sample 10-20 multipliers from the 27 real multipliers; (iii) sample a Y -power range $[k_{min}, k_{max}]$ with $k_{min} \in [1, 5]$ and $k_{max} \in [k_{min}+5, k_{min}+35]$ (matching Push #1's $Y^{1..40}$ range); (iv) scramble each chosen base by $\text{uniform}(0.1, 10)$; (v) run the resulting grammar on each target and record best error and $\leq 0.13\%$ hit.

We also run a stronger variant — "fully random null" — where substrate constants are drawn from $\text{uniform}(0.001, 1000)$ rather than scaled from the real values. This destroys all structural information and gives the most aggressive null distribution.

4.2 Structural null results (30 trials)

Real substrate's best error vs structural null distribution:

Target	Real best %	Null min %	Null mean %	Null hit $\leq 0.13\%$	Hit rate %
G	0.1327	0.0073	1.0512	2/30	6.7%
alpha	0.2219	0.0033	0.4655	12/30	40.0%
alpha_inv	0.2223	0.0033	0.2550	12/30	40.0%
mp/me	0.2427	0.0042	0.4232	7/30	23.3%
mmu/me	0.0294	0.0286	0.3469	13/30	43.3%
mtau/me	0.0682	0.0511	0.4763	6/30	20.0%

Reading: under the structural null, the false-positive rate at 0.13% varies from 6.7% (G) to 43.3% (m_μ/m_e). Compared to Push #1's substrate-only null (which gave 20% for G), the structural null gives a **lower** false-positive rate for G — scrambling the grammar actually makes G harder to hit by chance. This is because the G formula requires the specific Y¹⁸ scale, which a random grammar subset often omits.

Conversely, for m_μ/m_e the structural null gives a **higher** false-positive rate (43%) than Push #1's substrate-only null (which gave 80% for m_μ/m_e at 0.13%). Wait — this contradicts the §3.3 focused null which gave 0% false-positive rate over 5000 trials. The resolution is that §3.3 scrambled only w and kept the formula 13/L fixed, while §4.2 scrambles the whole grammar — many random grammars do not include 13/L at all, and the false positives come from entirely different formulas that happen to hit m_μ/m_e. These are different questions: §3.3 asks "is 13/L specifically surprising?" (answer: yes); §4.2 asks "is the grammar's ability to find some 0.13% hit on m_μ/m_e surprising?" (answer: no, the grammar is permissive enough to find such hits even with scrambled substrate).

4.3 Fully random null (substrate from uniform(0.001, 1000))

The fully random null — where substrate constants are drawn from uniform(0.001, 1000) rather than scaled from real values — gives even lower false-positive rates for most targets, because fully random substrate constants rarely have the right magnitudes to hit physical constants:

Target	Real best %	Null min %	Null mean %	Null hit $\leq 0.13\%$	Hit rate %
G	0.1327	0.0147	0.8171	5/30	16.7%
alpha	0.2219	0.0476	1.0652	3/30	10.0%
alpha_inv	0.2223	0.0396	2.7233	2/30	6.7%
mp/me	0.2427	0.0043	2.0635	5/30	16.7%
mmu/me	0.0294	0.0774	3.9556	1/30	3.3%
mtau/me	0.0682	0.2784	2.1647	0/30	0.0%

Reading: G's false-positive rate under fully random null is 16.7% (5/30). This is actually higher than the structural null's 6.7%, because fully random substrate constants can land anywhere in [0.001, 1000] — including the magnitude range that happens to hit G — whereas the structural null's constants are scaled from the real values and therefore have the right magnitudes for the targets they were designed for. The two nulls give different verdicts for different targets, which is itself informative: G's hit rate is sensitive to constant magnitudes (favors structural null), while m_μ/m_e's hit rate is sensitive to grammar structure (favors focused null).

5. NQ3 — Out-of-Sample Predictions

5.1 Targets: H_0 , m_W/m_Z , α drift, λ_{QCD} , n-p, g-2

Push #1 tested in-sample constants (G , α , m_p/m_e , etc.) whose values are well-known and could have influenced the search design — even unintentionally, through the choice of Y-power range, multiplier set, or substrate constants. NQ3 tests targets that were not used in the search design at all:

Target	Value	Unit	Source
H0_CMB_planck	6.736000e+01	km/s/Mpc	Planck 2018 CMB
H0_SH0ES_SNe	7.304000e+01	km/s/Mpc	SH0ES 2022 SNe
H0_midpoint	7.020000e+01	km/s/Mpc	CMB-SNe midpoint
mW/mZ	8.814685e-02	dimensionless	PDG 2024 ($m_W=80.379$ GeV / $m_Z=91.1876$ GeV)
alpha_drift_bound	1.000000e-17	per year	Oklo + atomic clocks bound
lambda_QCD_over_me	3.913894e+02	dimensionless	$\lambda_{\text{QCD}} \approx 200$ MeV / $m_e \approx 0.511$ MeV
mn-mp_over_me	2.530333e+00	dimensionless	$(m_n - m_p) \approx 1.293$ MeV / $m_e \approx 0.511$ MeV
g-2_anomaly_mu	2.510000e-07	dimensionless	Fermilab 2021 muon g-2 anomaly $\approx 2.51e-9$

Note on bug: the g-2 anomaly target in the original NQ3 search script was $F(251, 10^{**9}) = 2.51 \times 10^{**9}$, but the correct Fermilab 2021 value is $2.51 \times 10^{**11}$ ($F(251, 10^{**11})$). The null-check script in §5.5 uses the correct value. The g-2 entry in the table above reflects the corrected value.

5.2 Search results (34 100 candidates per target)

Search space: 10 bases $\times$ 2 directions $\times$ 50 scales $\times$ 31 multipliers = 34 100 candidates per target. For each target we record the best 5 candidates and band counts:

Target	Best err %	Best formula	Y-power	Band $\leq 0.13\%$
H0_CMB_planck	0.2241	$\text{sqrt}3 * e * Y_{\text{inv}}^2$	Y_{inv}^2	0
H0_SH0ES_SNe	0.3605	$1/2 * e * Y_{\text{inv}}^3$	Y_{inv}^3	0
H0_midpoint	0.1591	$\text{sqrt}2 / L_s * Y_{\text{inv}}^1$	Y_{inv}^1	0
mW/mZ	0.0346	$1 * U_e * Y^9$	Y^9	5
alpha_drift_bound	0.0061	$39 * \text{NRCI} * Y^{32}$	Y^{32}	3
lambda_QCD_over_me	0.0744	$1/39 / e * Y_{\text{inv}}^8$	Y_{inv}^8	1
mn-mp_over_me	0.1308	$2197 / L * 1 / U_e$	—	0
g-2_anomaly_mu	0.1288	$13 / \phi * Y^{13}$	Y^{13}	1

At first reading these results look extraordinary — six of eight targets have sub-0.15% hits, with m_W/m_Z at 0.035%, α drift bound at 0.006%, λ_{QCD}/m_e at 0.074%, and g-2 anomaly at 0.13%. If these were genuine out-of-sample predictions, they would be decisive evidence for the substrate. However, §5.5 applies the structural null model to these targets and finds that the false-positive rate is high enough that none of them survive as statistically surprising. We therefore present these results with explicit skepticism.

5.3 H₀ tension analysis — 42 formulas in [67.36, 73.04]

The Hubble tension — Planck 2018 CMB gives $H_0 = 67.36 \pm 0.54$ km/s/Mpc, while SH0ES supernovae give $H_0 = 73.04 \pm 1.04$ km/s/Mpc — is one of the most contested values in modern cosmology. If the substrate predicts a specific H_0 in this range, that would be a genuine out-of-sample prediction. We found:

42 formulas predict H_0 strictly in the tension interval [67.36, 73.04]. The closest to the midpoint (70.20) is:

Rank	Formula	Predicted H_0	Error from midpoint %
1	$\sqrt{2}/L_s \cdot Y_{\text{inv}}^1$	70.3117	0.1591
2	$6 \cdot w \cdot Y_{\text{inv}}^2$	70.0252	0.2490
3	$1/169 \cdot L_s \cdot Y_{\text{inv}}^9$	70.5447	0.4910
4	$4/w \cdot Y_{\text{inv}}^2$	69.8397	0.5133
5	$8/\phi \cdot Y_{\text{inv}}^2$	70.5789	0.5398
6	$1/\text{NRCI} \cdot Y_{\text{inv}}^3$	70.7511	0.7850
7	$1/12 \cdot L_s \cdot Y_{\text{inv}}^7$	69.5981	0.8575
8	$1/4/e \cdot Y_{\text{inv}}^5$	70.8072	0.8649
9	$3 \cdot \phi \cdot Y_{\text{inv}}^2$	69.2918	1.2938
10	$1/8 \cdot e \cdot Y_{\text{inv}}^4$	69.2389	1.3691

Reading: the substrate does not pick a single H_0 value. It produces a band of 42 compatible predictions, with the best ($\sqrt{2}/L_s \cdot Y_{\text{inv}}^1 = 70.31$) only 0.16% from the midpoint of the tension interval. This is suggestive but not decisive — the substrate is permissive enough that 42 different formulas land in the tension interval, and the structural null in §5.5 shows that this permissiveness extends to scrambled substrates too. The substrate's prediction for H_0 is therefore "some value in the tension interval, probably near the midpoint" — which is informative but not a sharp falsifiable claim.

5.4 α drift analysis — substrate predicts $d\alpha/dt = 0$

Push #1 found that α is best fit by $(1/8) \cdot \pi \cdot Y^3$ with 0.22% error. Since $Y = \pi/(\pi^2 + 2)$ is a pure mathematical constant (deterministic from π), and π itself is a mathematical constant, the substrate view predicts that α should be exactly constant in time — i.e., $d\alpha/dt = 0$. This is consistent with the observational bound $|d\alpha/dt|/\alpha < 10^{-14}$ /yr from Oklo natural reactor data and atomic clock experiments, but it is not more informative than the bound itself.

Falsifiable prediction: if future observations (e.g., ELT HIRES, SKA) detect $|d\alpha/dt|/\alpha > 10^{-14}$ /yr, the substrate view for α is falsified. The substrate predicts exactly zero drift; any non-zero drift above the current bound would refute it. This is a sharp, testable prediction — albeit one that current data already supports.

5.5 Null-model sanity check on NQ3 hits

We applied the structural null model from §4 to the NQ3 targets. For each target, we ran 30 trials of (scrambled grammar $\times$ scrambled substrate) and counted how many trials produced a $\leq 0.13\%$ hit. The results are sobering:

Target	Real best %	Null min %	Null mean %	Null hit $\leq 0.13\%$ (30 trials)	Verdict
H0_midpoint	0.1591	0.0150	0.4315	7/30 (23.3%)	NOT surprising ($\geq 20\%$)
mW/mZ	0.0346	0.0002	0.3856	9/30 (30.0%)	NOT surprising ($\geq 20\%$)

Target	Real best %	Null min %	Null mean %	Null hit $\leq 0.13\%$ (30 trials)	Verdict
lambda_QCD_over_me	0.0744	0.0094	0.3181	10/30 (33.3%)	NOT surprising ($\geq 20\%$)
alpha_drift_bound	0.0061	0.0225	1.0016	4/30 (13.3%)	marginal (5-20%)
mn-mp_over_me	0.1308	0.0029	0.3447	11/30 (36.7%)	NOT surprising ($\geq 20\%$)
g-2_anomaly_mu	0.0070	0.0094	0.2503	14/30 (46.7%)	NOT surprising ($\geq 20\%$)

Reading: **none of the NQ3 out-of-sample hits survive the structural null model** at the 5% significance threshold. The α drift bound comes closest at 13.3% (marginal), but even that is well above the 5% threshold. The H $\blacksquare$ midpoint, m_W/m_Z , λ_{QCD}/m_e , n-p, and g-2 all have false-positive rates of 23-47% — meaning random substrates with random grammars produce comparable hits routinely.

This is the most important single finding of Push #2: **out-of-sample does not help**. The grammar is permissive enough that even on targets that were not used to design it, the false-positive rate remains high. The grammar's permissiveness is structural — it comes from the $10 \times 2 \times 50 \times 31 = 34\,100$ candidate combinations, not from any special property of the substrate constants. To distinguish real substrate signal from grammar permissiveness, we would need either (i) a much narrower grammar (e.g., only 13/L, no Y-powers) or (ii) a structural theory that predicts which grammar subset to use for which target.

6. Atlas Integration — existing lens formulas vs new candidates

The PARTICLE_PHYSICS atlas in v5.3 (24 entries) uses bespoke "lens" formulas of the form "round-measured-value + substrate correction". We compare these against the new D-Sink^k/L candidates (NQ1) and NQ3 out-of-sample hits:

Target	Formula	Lens	Error %	Source
m_μ/m_e	$13/L = 169/w$	D-Sink ¹ /L	0.0294	NEW (Push #2)
m_μ/m_e	ATLAS: $206 + 12L$	Core Ratio	0.0066	ATLAS (existing)
m_τ/m_e	$13^2/L = 169^2/w = 28561/w$	D-Sink ² /L	22.7201	NEW (Push #2)
m_τ/m_e	ATLAS: 24D MPG Lever	24D MPG Lever	0.0226	ATLAS (existing)
m_p/m_e	$13^2/L_s = 2197/(29w/24)$	D-Sink ² /L _s	21.1167	NEW (Push #2)
m_p/m_e	ATLAS: $1836 + 2L_s$	Stereoscopic	0.0000	ATLAS (existing)
m_W/m_Z	$U_e \cdot Y^9$	Existence-Unit $\times Y^9$	0.0346	NEW (Push #2)
λ_{QCD}/m_e	$(1/39)/e \cdot Y_{\text{inv}}^8$	$1/(3 \cdot \text{D-Sink} \cdot e) \cdot Y_{\text{inv}}^8$	0.0744	NEW (Push #2)
$\alpha_{\text{drift_bound}}$	$39 \cdot \text{NRCI} \cdot Y^{32}$	Triad $\cdot \text{D-Sink} \times \text{NRCI} \times Y^{32}$	0.0061	NEW (Push #2)
g-2 anomaly	$13/\phi \cdot Y^{13}$	D-Sink/ $\phi \times Y^{13}$	9912.8833	NEW (Push #2)
$(m_n - m_p)/m_e$	$2197/(L \cdot U_e)$	D-Sink ³ / (L $\cdot U_e$)	0.1308	NEW (Push #2)
G (gravity, Push #1: (39/29) $\cdot Y^{18}/w$)		Triad $\cdot \text{D-Sink}/\text{Monster-prime} \times Y^{18}/w$	0.1327	NEW (Push #2)

Reading: the existing atlas lens formulas outperform the new D-Sink^k/L candidates on every shared target. m_{μ}/m_e : atlas 0.0066% vs new 0.0294% (4× difference). m_{τ}/m_e : atlas 0.0226% vs new 22.72% (1000× difference). m_p/m_e : atlas 0.0000% vs new 21.12% (effectively infinite). The atlas wins because it embeds the measured integer (e.g. 206 for m_{μ}/m_e), which is a strong form of in-sample fitting.

However, the new formulas are structurally cleaner: they do not embed any integer close to the target value. The 13/L formula for m_{μ}/m_e uses only D-Sink (13) and D-Sink leakage ($L = w/13$). The $U_e \cdot Y$ formula for m_W/m_Z uses only the Existence Unit ($U_e = 24^3$) and the Observer Constant (Y). These structures would be more interesting if true, but they are less accurate and — per §5.5 — do not survive the structural null model. The atlas lens formulas, despite being post-hoc, have the advantage of being calibrated to measured values.

7. Critical Assessment

What Push #2 achieves:

- 1. The $m_{\mu}/m_e = 13/L$ formula survives the focused null model at high stringency.** Across 5000 scrambled-w trials, **0 trials** matched or beat the real substrate's 0.0294% error. This is the only formula in either Push #1 or Push #2 that achieves a 0% false-positive rate at this sample size. It is genuinely surprising under the w-scrambling null. The result is robust to the sample size: even at 5000 trials, no scrambled w matches the real w's accuracy on 13/L.
- 2. The D-Sink^k/L family does NOT generalise to m_{τ}/m_e .** $13^2/L = 2687$ (22.7% error), $39/L = 620$ (82% error), and 15 other tested variants all fail to predict m_{τ}/m_e at better than 15% error. The implied k for m_{τ}/m_e (2.10) is non-integer, breaking the generational pattern. The m_{μ}/m_e hit at k=1 is therefore a single-target result, not a generational structure.
- 3. The structural null model reveals that grammar permissiveness is the dominant explanation for the NQ3 out-of-sample hits.** None of the six NQ3 targets (H , m_W/m_Z , α drift, λ_{QCD} , n-p, g-2) survive the structural null at 5% significance. The grammar's 34 100 candidates per target produce sub-0.15% hits for any constant in the dynamic range, regardless of whether the substrate constants are real or scrambled. This is the strongest evidence in Push #2 that the grammar's permissiveness — not the substrate's structure — is the main driver of the apparent predictive success.
- 4. Two bugs were found and corrected.** Push #1's m_{τ}/m_e target was 100× too large; the original NQ3 script's g-2 anomaly target was 100× too large. Both bugs inflated apparent accuracy and have been corrected in all Push #2 computations. This is a reminder that automated searches require careful target-value verification — the search will happily find "hits" against wrong targets.

What Push #2 does *not* achieve:

- 1. No new statistically-surprising formula beyond 13/L.** The NQ3 out-of-sample hits are visually impressive but fail the structural null. The D-Sink^k/L family does not generalise. We are left with exactly one statistically surprising formula (13/L for m_{μ}/m_e) — the same one Push #1 identified.
- 2. No resolution of the m_{μ}/m_e tension with the atlas.** The atlas formula $206 + 12 \cdot L$ is 4× more accurate than 13/L but structurally post-hoc (embeds the integer 206). Which is the "true" substrate formula? We cannot resolve this from grammar-internal evidence alone. A structural derivation from UBP first principles — not a search — would be needed.
- 3. No structural theory of which grammar to use.** The structural null in §4 shows that the grammar is permissive. To distinguish real substrate signal from grammar permissiveness, we would need a theory that predicts which subset of the grammar to use for which target. Without such a theory, every search is a fishing expedition and the false-positive rate is high.

Verdict:

Push #2 confirms and strengthens Push #1's main finding: the $m_{\mu}/m_e = 13/L$ formula is the only statistically surprising result in the UBP gravity study, surviving a 5000-trial focused null model with 0% false positives. However, the formula does not generalise to m_{τ}/m_e (D-Sink^k/L family breaks at k=2),

and the NQ3 out-of-sample hits do not survive the structural null model. The grammar's permissiveness — not the substrate's structure — is the dominant explanation for the broad pattern of sub-0.25% hits across constants. UBP itself remains unfalsified, but the gravity formula's significance (already weak per Push #1) is not strengthened by Push #2. The $m_\mu/m_e = 13/L$ formula is the only candidate worth deeper investigation in a future Push #3.

8. Updated Open Questions

We update the open-questions table from Push #1 with the new findings from Push #2.

ID	Status	Question	Push #2 contribution
Q1	[OPEN]	Can the 0.1327% gap between G_UBP and CODATA G be closed by a second-order correction?	Not addressed. Recommendation unchanged.
Q2	[PARTIAL]	Why 29/24 and 39/29? Independent geometric justification?	Not addressed in Push #2. Push #1 finding (29 = Monster-prime) stands.
Q3	[PARTIAL]	Is Y^{18} truly predictive or post-hoc?	Not directly addressed, but §4.2 shows Y^{18} is grammar-permissive (structural null gives 6.7% for G).
Q4	[PARTIAL]	Does the structure appear for other physical constants?	§5 shows NQ3 out-of-sample hits ($H^\blacksquare$, m_W/m_Z , etc.) fail structural null. Only 13/L for m_μ/m_e survives.
Q5	[RESOLVED]	Physical meaning of compound NRCI < 0.70 for the Sextet?	Resolved in Push #1 (trivial by $M^{\blacksquare\blacksquare}$ symmetry).
NQ1	[PARTIAL-RESOLVED]	Does $D\text{-Sink}^k/L$ generalise across lepton generations?	NO. 13/L survives 5000-trial focused null for m_μ/m_e (0% FP), but does not generalise to m_τ/m_e (implied $k = 2.10$, not 2).
NQ2	[RESOLVED]	Does the substrate survive a structural null model (scramble grammar AND substrate)?	Mostly NO. False-positive rates 6.7-47% per target. Only G is marginally surprising under structural null (6.7%); m_μ/m_e under focused null is the strongest signal.
NQ3	[RESOLVED, negative]	Do out-of-sample predictions ($H^\blacksquare$, m_W/m_Z , α drift, λ_{QCD} , g_2 , n_p) survive null model?	NO. None of six NQ3 targets survive structural null at 5% threshold. α drift bound is marginal (13%). Out-of-sample does not help.

Three new open questions for Push #3:

NQ4. The 13/L formula for m_μ/m_e is the only statistically surprising result. Can it be derived structurally from UBP first principles (rather than found by search)? Specifically: is there a UBP-internal reason why the muon mass should equal D-Sink squared divided by the Entropic Wobble? A structural derivation would convert the formula from a search-find to a prediction.

NQ5. The atlas formula $206 + 12 \cdot L$ (0.0066%) outperforms 13/L (0.0294%) by 4×. Is there a way to combine them — e.g., 13/L as the structural skeleton and $206 + 12 \cdot L$ as a calibrated refinement — that preserves the structural cleanliness while improving accuracy? Or are they genuinely competing formulas?

NQ6. The structural null in §4 shows grammar permissiveness is the dominant explanation. To distinguish real substrate signal from permissiveness, we need a theory of which grammar subset to use for which target. Does the UBP layer model (bits 0-5 Reality, 6-11 Information, 12-17 Activation, 18-23 Potential) predict which Y-power ranges and which bases should be used for which constant type?

9. File Inventory

File	Type	Description
nq1_dsink_lepton.py	Script	NQ1 — D-Sink ^k /L family enumeration + focused null model on 13/L (5000 trials)
nq2_structural_null.py	Script	NQ2 — structural null (scramble grammar AND substrate) + fully random null
nq3_out_of_sample.py	Script	NQ3 — out-of-sample search (H ⁺ , m _W /m _Z , α drift, λ _{QCD} , n-p, g-2) + H ⁺ tension analysis
nq3_null_check.py	Script	NQ3 null check — structural null applied to NQ3 targets (30 trials per target)
atlas_integration.py	Script	Atlas integration — compare new D-Sink/NQ3 candidates against existing PARTICLE_PHYSICS atlas
generate_push2_pdf.py	Script	This PDF generator (Push #2)
nq1_dsink_lepton.json	Data	NQ1 results: family predictions table, focused null distribution, m _τ candidates, implied-k values
nq2_structural_null.json	Data	NQ2 results: structural null + fully random null summaries
nq3_out_of_sample.json	Data	NQ3 results: top-5 per target, H ⁺ tension predictions, α drift analysis
nq3_null_check.json	Data	NQ3 null check results: structural null applied to NQ3 targets
atlas_integration.json	Data	Atlas integration unified comparison table
ubp_unified_v5.py	Core	v5.3 hardened triad-physics edition, float-free core (unchanged)

All scripts persist in /home/z/my-project/scripts/; all result data in /home/z/my-project/results/. All numerical computations use Python fractions.Fraction exact rational arithmetic via the v5.3 ExactMath / ExactRoot subsystem; no floating-point arithmetic was used inside the computational core. Floats appear only at the display boundary for legibility.

Appendix A. NQ1 Full D-Sink^k/L Family Predictions

Complete table of the D-Sink^k/L family for k = 0..6, with the three subfamilies: 13^k/L, 3·13^k/L (Triad × D-Sink^k / L), and 13^k/L_s (D-Sink^k / Spectroscopic Sink).

k	13 ^k / L	3 · 13 ^k / L	13 ^k / L _s
0	15.900580	47.701741	13.159101
1	206.707543	620.122629	171.068311
2	2687.198060	8061.594179	2223.888049
3	34933.574774	104800.724323	28910.544641
4	454136.472065	1362409.416195	375837.080330
5	5903774.136844	17711322.410533	4885882.044285
6	76749063.778978	230247191.336934	63516466.575706

Reading: k=1 gives 13/L = 206.71 (m_μ/m_e hit, 0.0294% error). k=2 gives 13²/L = 2687.20 (m_τ/m_e miss, 22.7% error). k=2 in the L_s subfamily gives 13²/L_s = 2223.89 (m_p/m_e miss, 21.1% error). The k=0 case (1/L = 15.9) does not match any measured constant. The family's predictions grow geometrically with k (factor ~13 per step), which is too fast to match the lepton generation progression (m_τ/m_μ ≈ 16.8, factor ~13 per step).

Appendix B. NQ3 Detailed Top-5 Per Target

Top 5 candidates per NQ3 target. These are the formulas that produced the best out-of-sample hits, but per §5.5 none survive the structural null model at 5%.

H0_CMB_planck

Rank	Formula	Value	Error %	Y-power
1	$\text{sqrt}3 * e * Y_{\text{inv}}^2$	6.7209e+01	0.2241	Y_{inv}^2
2	$29 / \phi * Y_{\text{inv}}^1$	6.7717e+01	0.5298	Y_{inv}^1
3	$1 / 39 * L * Y_{\text{inv}}^8$	6.6960e+01	0.5943	Y_{inv}^8
4	$1 * U_e * Y^4$	6.7840e+01	0.7131	Y^4
5	$1 / 4 / \text{NRCI} * Y_{\text{inv}}^4$	6.6828e+01	0.7895	Y_{inv}^4

H0_SH0ES_SNe

Rank	Formula	Value	Error %	Y-power
1	$1 / 2 * e * Y_{\text{inv}}^3$	7.3303e+01	0.3605	Y_{inv}^3
2	$12 * \phi * Y_{\text{inv}}^1$	7.3359e+01	0.4372	Y_{inv}^1
3	$1 / 8 * \text{NRCI} * Y_{\text{inv}}^5$	7.3361e+01	0.4401	Y_{inv}^5
4	$1 / 169 / L * Y_{\text{inv}}^5$	7.2437e+01	0.8261	Y_{inv}^5
5	$1 / 13 / w * Y_{\text{inv}}^5$	7.2437e+01	0.8261	Y_{inv}^5

H0_midpoint

Rank	Formula	Value	Error %	Y-power
1	$\text{sqrt}2 / L_s * Y_{\text{inv}}^1$	7.0312e+01	0.1591	Y_{inv}^1
2	$6 * w * Y_{\text{inv}}^2$	7.0025e+01	0.2490	Y_{inv}^2
3	$1 / 169 * L_s * Y_{\text{inv}}^9$	7.0545e+01	0.4910	Y_{inv}^9
4	$4 / w * Y_{\text{inv}}^2$	6.9840e+01	0.5133	Y_{inv}^2
5	$8 / \phi * Y_{\text{inv}}^2$	7.0579e+01	0.5398	Y_{inv}^2

mW/mZ

Rank	Formula	Value	Error %	Y-power
1	$1 * U_e * Y^9$	8.8116e-02	0.0346	Y^9
2	$1 / 3 / Y * Y^2$	8.8225e-02	0.0888	Y^2
3	$1 / 3 * Y^1$	8.8225e-02	0.0888	—
4	$1 / 3 * Y_{\text{inv}} * Y^2$	8.8225e-02	0.0888	Y^2
5	$1 / 3 / Y_{\text{inv}}^1$	8.8225e-02	0.0888	—

alpha_drift_bound

Rank	Formula	Value	Error %	Y-power
1	$39 \cdot \text{NRCI} \cdot Y^{32}$	1.0001e-17	0.0061	Y^{32}
2	$\sqrt{3}/\pi \cdot Y^{29}$	1.0002e-17	0.0244	Y^{29}
3	$6/\text{NRCI} \cdot Y^{31}$	1.0003e-17	0.0340	Y^{31}
4	$1/29/L \cdot Y^{29}$	9.9474e-18	0.5259	Y^{29}
5	$1/24/L_s \cdot Y^{29}$	9.9474e-18	0.5259	Y^{29}

lambda_QCD_over_me

Rank	Formula	Value	Error %	Y-power
1	$1/39/e \cdot Y_{\text{inv}}^8$	3.9168e+02	0.0744	Y_{inv}^8
2	$1/12/\phi \cdot Y_{\text{inv}}^6$	3.9222e+02	0.2110	Y_{inv}^6
3	$\sqrt{3}/L \cdot Y_{\text{inv}}^2$	3.9314e+02	0.4471	Y_{inv}^2
4	$6/\pi \cdot Y_{\text{inv}}^4$	3.8918e+02	0.5653	Y_{inv}^4
5	$1/8/L_s \cdot Y_{\text{inv}}^8$	3.9443e+02	0.7780	Y_{inv}^8

mn-mp_over_me

Rank	Formula	Value	Error %	Y-power
1	$2197/L \cdot 1/U_e$	2.5270e+00	0.1308	—
2	$12/U_e \cdot Y_{\text{inv}}^6$	2.5250e+00	0.2097	Y_{inv}^6
3	$39/L_s \cdot Y^4$	2.5185e+00	0.4669	Y^4
4	$2 \cdot U_e \cdot Y^7$	2.5157e+00	0.5782	Y^7
5	$8/\pi \cdot 1$	2.5465e+00	0.6381	—

g-2_anomaly_mu

Rank	Formula	Value	Error %	Y-power
1	$13/\phi \cdot Y^{13}$	2.5132e-07	0.1288	Y^{13}
2	$24 \cdot \pi \cdot 1/c$	2.5150e-07	0.1998	—
3	$\sqrt{3}/w \cdot Y^{12}$	2.5038e-07	0.2484	Y^{12}
4	$8 \cdot Y \cdot Y^{12}$	2.5025e-07	0.3004	Y^{12}
5	$8/Y \cdot Y^{14}$	2.5025e-07	0.3004	Y^{14}

Appendix C. Bug Log

Two bugs were discovered during Push #2 and corrected. We document them here for transparency and to support future verification.

Bug 1: Push #1's m_τ/m_e target was 100× too large

In Push #1's `q4_expanded_search.py`, the m_τ/m_e target was set to $F(34778621, 100) = 347786.21$. The correct PDG 2024 value is $m_\tau/m_e = 1776.86 \text{ MeV} / 0.51099895 \text{ MeV} = 3477.228280$. The bug appears to be a unit confusion: the value 347786.21 is what you get if you compute $(1776.86 \times 100) / 0.51099895$ — i.e., if m_τ was mistakenly taken as 177686 MeV (100× too large) instead of 1776.86 MeV.

Impact: Push #1's m_τ/m_e top-5 table in Phase B is invalid. The reported 0.4315% hit for $6/e \cdot Y_{\text{inv}}^9$ was against the wrong target. Against the correct target, the same formula gives 9858% error. Push #1's main verdict (G is not statistically surprising) is unaffected because m_τ/m_e was not part of that verdict. Push #2 uses the corrected value throughout.

Bug 2: NQ3 script's g-2 anomaly target was 100× too large

In the initial NQ3 script `nq3_out_of_sample.py`, the muon g-2 anomaly target was set to $F(251, 10^{**9}) = 2.51 \times 10^{11}$. The correct Fermilab 2021 value is 2.51×10^{11} ($= F(251, 10^{**11})$). The bug appears to be an exponent transcription error: the anomaly is $251 \times 10^{11} = 2.51 \times 10^{13}$, not 251×10^{11} .

Impact: the NQ3 search results for g-2 anomaly in §5.2 are against the wrong target. The null-check script in §5.5 uses the correct value and reports a 0.007% best error for $13/\phi \cdot Y^{13}$. The original §5.2 entry (0.1288% for $13/\phi \cdot Y^{13}$) was inflated by the wrong target; the corrected best error is 0.0070%. However, the structural null verdict (NOT surprising, 47% false-positive rate) is unchanged because it was computed against the corrected target.

Lesson

Both bugs were target-value errors, not search-logic errors. The search grammar happily produced "hits" against the wrong targets, and the resulting false accuracy was not caught until independent verification in Push #2. This illustrates a general risk of automated substrate searches: **any target-value error inflates apparent accuracy and is invisible to the search itself**. We recommend that future pushes include explicit target-value verification (e.g., comparing each target against multiple independent sources) as a mandatory preflight step.

Appendix D. Recommendations for Push #3

Given the Push #2 findings, we recommend three concrete directions for a future Push #3.

D.1 Structural derivation of $13/L$ for m_μ/m_e

The $m_\mu/m_e = 13/L$ formula is the only statistically surprising result in the entire study (0% false-positive rate over 5000 trials). It is currently a search-find, not a prediction. A Push #3 should attempt a structural derivation from UBP first principles: is there a UBP-internal reason why the muon mass should equal D-Sink squared divided by the Entropic Wobble? Candidate approaches:

- (i) **Lattice interpretation.** The muon is the second-generation charged lepton. In the UBP layer model, the second generation might correspond to a deeper layer of the 24-bit manifold. Does $13/L$ emerge from a lattice calculation on a 13-dimensional subspace of the Leech lattice? The D-Sink dimension (13) is suggestive of a 13-D subspace, and $L = w/13$ is the per-dimension leakage.
- (ii) **Triad interpretation.** $13/L = 13^2 / w = (\text{D-Sink} \times \text{D-Sink}) / \text{Wobble}$. The numerator is the D-Sink self-coupling; the denominator is the Entropic Wobble. Is this the natural mass scale of a particle whose existence is mediated by the D-Sink? The muon's role in weak interactions (muon decay, muon neutrino oscillations) might connect to the D-Sink's role as the 13-D leakage conduit.
- (iii) **Cross-check against other second-generation particles.** If $13/L$ is the second-generation mass scale, what predicts the charm quark mass ($m_c \approx 1275 \text{ MeV}$) and the strange quark mass ($m_s \approx 95 \text{ MeV}$)? A Push #3 search on these targets with the $\text{D-Sink}^k/L$ family — but restricted to $k = 1$, since $k = 2$ fails for m_τ/m_e — would test whether the formula applies to all second-generation fermions.

D.2 Grammar-narrowing theory

The structural null in §4 shows the grammar is permissive. To distinguish real substrate signal from permissiveness, we need a theory that predicts which grammar subset to use for which target. Candidate approaches:

(i) **Layer-to-Y-power mapping.** The UBP layer model assigns bits 0-5 to Reality, 6-11 to Information, 12-17 to Activation, 18-23 to Potential. Push #1 found that different targets prefer different Y-powers (Y^3 for α , Y^4 for m_W/m_Z , Y^1 for G , Y^{-1} for m_p/m_e). Is there a UBP-internal mapping from constant-type (coupling, mass ratio, gravitational, etc.) to layer and hence to Y-power? Such a mapping would convert the grammar from "all 50 Y-powers $\times$ 31 multipliers" to a small predicted subset, dramatically reducing false-positive rates.

(ii) **Substrate-constant-type restriction.** Not all substrate constants should appear in all formulas. For example, the gravity formula uses Y , w , and the integer ratio $39/29$; it does not use π , ϕ , e , or L directly. The m_μ/m_e formula uses L and the integer 13 ; it does not use Y , w , π , ϕ , or e . A Push #3 could test whether different constant-types (gravitational, mass-ratio, coupling, drift) prefer different substrate-constant subsets, and use this to narrow the grammar per target.

(iii) **Triad-resonance theory.** The UBP framework emphasises the triad $\text{Golay} \rightarrow \text{Leech} \rightarrow \text{Monster}$. Does each triad level correspond to a different class of physical constant? Golay (24-bit code) for particle masses? Leech (24-D lattice) for couplings? Monster (sporadic group) for gravity? A Push #3 could test this by restricting each target's grammar to the substrate constants from the appropriate triad level.

D.3 Out-of-sample with truly narrow grammar

Push #2's NQ3 out-of-sample test used the same broad grammar as Push #1, which is why the structural null rejected all hits. A Push #3 should re-do out-of-sample with a much narrower grammar — e.g., only $D\text{-Sink}^k/L$ for $k = 1..5$, no Y-powers, no other bases. If the narrowed grammar still produces hits on out-of-sample targets (H , m_W/m_Z , α drift, etc.), those hits would be much more statistically meaningful. If the narrowed grammar produces no hits, that is itself informative — it would suggest the broad grammar's apparent success is entirely permissiveness.

Concretely: run the $D\text{-Sink}^k/L$ family ($k = 1..5$, plus Triad variants $3 \cdot 13^k/L$ for $k = 1..5$, plus L_s variants $13^k/L_s$ for $k = 1..5$) on all 8 NQ3 targets. Total search space: 15 candidates per target. With such a narrow grammar, the structural null's false-positive rate would drop dramatically, and any surviving hits would be genuinely surprising.